



An undergraduate overview on quantum computing

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(Received January 25, 2024, Published online February 6, 2024)

This paper is a literature review on quantum computing. It presents a comprehensive overview of the history of quantum computing, including its origins in quantum mechanics. It also presents an overview on how quantum computing works, from qubits to quantum gates and circuits. Three subfields of quantum computing (quantum information science, quantum cryptography, and quantum machine learning) are discussed to provide more insight into the real-world applications of quantum computing. Lastly, the paper discusses the future of quantum computing and current research being done to further the field.

I. INTRODUCTION

Within the last decade, quantum computing, as a topic of conversation, has greatly increased in popularity; it has made its way into conversations in every setting. People are excited about a new technology that they are told can solve many never-been-done-before problems and have a great impact on society. Because of its increase in popularity, it is beneficial to know how it works and what its real-world applications are.

Quantum computing is simply computing that uses the principles of quantum mechanics to perform calculations. Therefore, it would not have been a thing if it were not for the origin of quantum mechanics. In 1905, Albert Einstein published his findings on the photoelectric effect, proving the quantization of particles [1]. This was the beginning of quantum mechanics, a field that describes the nature of the universe at an atomic level. The term “quantum mechanics” was later coined by Max Born. It was only until 1980 that Paul Benioff took the foundation of quantum mechanics and published the first paper describing the possibility of a quantum computer. After Benioff’s work, research into creating a quantum computer increased, and many quantum initiatives were created to support the process. The research into the field is still ongoing as many scientists are determined to create a quantum computer that can solve problems that a classical computer cannot.

This paper will describe the workings of a quantum computer, specifically the workings of qubits to build quantum gates and circuits, to provide insight into the relative problems that quantum computers can solve. Relevant subfields in quantum computing will be discussed, as well as the future trajectory of the field as a whole, to further show ways quantum computers can solve problems differently than classical computers and have a great impact on the world.

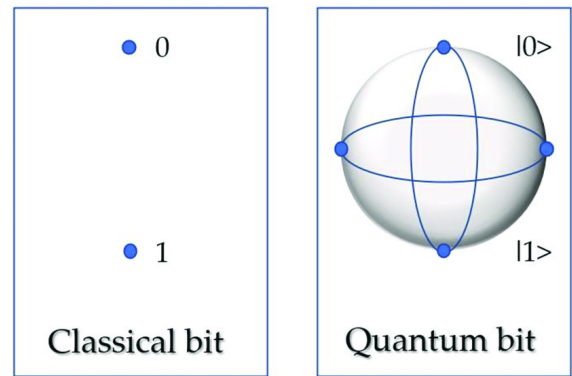


FIG. 1. A classical bit can only be in a state of 0 or 1 as shown by the diagram on the left. A quantum bit can also exist in a state of 0 or 1, but it can also exist in a combination of these states. All possible combinations are represented as the surface of the sphere. [4]

II. QUBITS

Classical computers are built around the idea of a binary bit: a basic unit of information that stores a value of 0 or 1. These bits can be physically represented as voltages where 0 is 0 V and 1 is a small voltage such as 5 V [2]. These bits can be manipulated via the use of classical gates which are physically electrical circuits. Classical gates perform logical operations on bits, such as AND, OR, and NOT, in order to process data.

Quantum computers, on the other hand, are built around the idea of a quantum bit or qubit: a unit of information that can be in a 0 or 1 state but also in a combination of these states [3]. Qubits are physically represented with electrons, ions, or photons, and they can be manipulated using quantum gates. An overview of quantum gates will be covered in the next section. A visual representation of the difference between classical bits and qubits is shown in fig. 1.

A. Quantum Properties

There are various types of qubits, but before discussing this, we must first discuss two important quantum mechanics properties from the perspective of a qubit: superposition and entanglement. First, superposition is the property that a qubit can exist in multiple states at the same time. This property comes from quantum mechanics and has to do with the wavelike nature of particles [5]. Quantum waves are governed mathematically by the Schrodinger wave equation which describes the probability distributions of a particle. These probability distributions can describe the likelihood a particle is residing in a specific location. They can also describe the probability a particle has a specific speed. Either way, because there are only probabilities that a particle has a specific position or speed, we say the particle is in a superposition of all possible states until it is measured. For the case of the qubit, this means the qubit can exist in multiple combinations of the 1 or 0 states at the same time. Because it is said to be in multiple states, calculations can be done on all possible states at the same time as opposed to a single calculation at a time for a classical bit.

Quantum computers also utilize the property of entanglement [6]. Quantum entanglement describes a correlation between two quantum systems that are separated. The property of entanglement arises from fundamental physics conservation laws. Entangled particles are typically generated by breaking one particle down into two [7]. Certain properties like momentum, spin, and angular momentum are conserved between the two particles. This means if the state of one particle is measured, the state of the entangled particle will also be known due to laws of conservation. For a quantum computer, entanglement allows for a calculation in one qubit to result in a calculation of the entangled qubit. This is known as parallel computing and it expedites the calculation process.

A graphical representation of a qubit's state is shown in the Bloch sphere in fig. 2 [8]. The north and south poles of the sphere represent the 1 and 0 states. In the diagram, these states are represented in Dirac notation as $|0\rangle$ and $|1\rangle$ [9]. The surface of the sphere represents a superposition state.

B. Types of Qubits

There are many types of qubits, each assigning states in different ways [11]. Superconducting qubits, one of the most prevalent types, assign states based on the clockwise or counterclockwise direction of a superconductor's current. A spin qubit uses the naturally occurring spin states of up or down. Trapped ion qubits assign states based on electrons being in either a high or low energy level in an atom or ion. Photonic qubits use a photon's polarization, either horizontal or vertical, as its state.

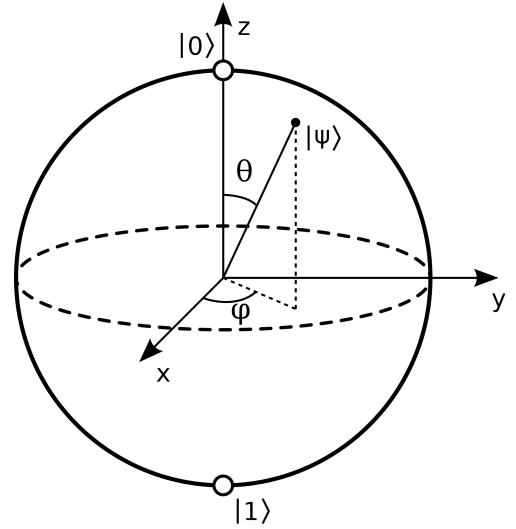


FIG. 2. A Bloch sphere shows the possible superposition of states for a quantum bit. The 1 and 0 states are represented in Dirac notation as $|0\rangle$ and $|1\rangle$. [10]

The states for photonic qubits can also be described by the path a photon takes or the time a photon takes to arrive at a specific point. Many other types of qubits can be created based on other physics properties. They can be based on a particle's topological properties or other applications of electron spin and nuclear spin states.

III. QUANTUM GATES

Just like qubits are the quantum computer representation of classical bits, quantum logic gates are the quantum computer representation of classical logic gates [12]. A classical logic gate processes information by producing an output from two inputs and creates the foundation of circuits that are used to perform operations within the computer. For a classical computer, logic gates block or allow the flow of electricity through the use of transistors. Flowing electricity represents a 1 bit and blocked electricity represents a 0 bit. The type of gate determines which bits are blocked or allowed, and in doing so, generates specific output with specific voltages. When the gates are arranged in a circuit, certain operations can be performed within the computer. Quantum logic gates operate under the same principle as classical logic gates, but instead of changing the resulting outcome voltage, quantum gates change the quantum state of qubits [13].

A. Types of Quantum Gates

Common quantum gates include Pauli gates which are the X, Y, and Z gates. The X gate is also known as the NOT gate in quantum computing. When Pauli gates are applied to qubits, their state is transformed to a state

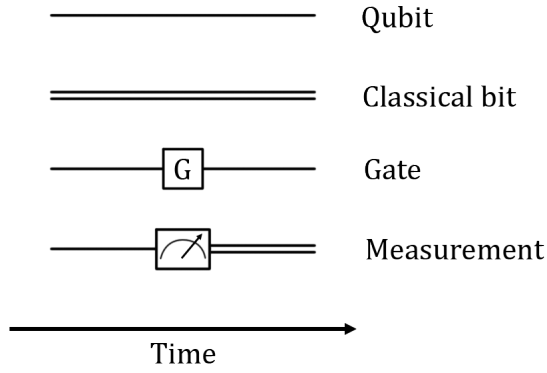


FIG. 3. The various components of a quantum circuit are shown. [17]

that is a 180-degree rotation around the corresponding x , y , or z axis on the Bloch sphere. Pauli gates are especially important because they are used in error correction. Typical errors include a sign flip or a phase flip. To correct these errors, Pauli gates are applied to return the qubit to its corrected state.

A Hadamard gate is another common gate that creates an equivalent superposition of the ($|0\rangle$ or $|1\rangle$) state of a qubit [14]. Because the state is now a superposition, the properties of superposition can be applied and multiple computations can happen simultaneously for the same qubit. The Hadamard gate is also self-inverse, meaning it can return qubits to their original state when applied twice. This makes it useful for error correction. There are also various gates, such as the S and T gates, that result in a phase shift of the state by a certain degree. The phase shift also creates a superposition of states.

The gates listed previously are all single-qubit gates. This means they can only act on one qubit at a time. There are multi-qubit gates that can act on multiple qubits [15]. These gates include the CNOT (Controlled-NOT) gate, which holds one qubit as a control and changes the state of the other qubit given the control is in a certain state. Another gate is the SWAP gate which swaps the state of two qubits. The multi-qubit gates and single-qubit gates are used together to create desired quantum states.

All of these states used together can build quantum circuits [16]. These circuits run the operations of the quantum computer. When creating circuit diagrams, the gates are represented as boxes and the circuit flows from left to right. The components of a quantum circuit diagram are shown in fig. 3. An example of a quantum circuit diagram is shown in fig. 4.

B. Putting it All Together

A quantum computer works by using specific quantum gates to create a desired circuit, and it works because of the quantum mechanics property of interference [19].

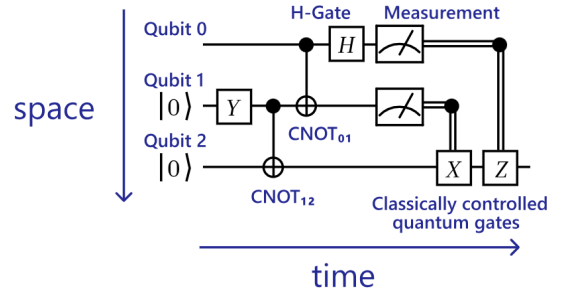


FIG. 4. A quantum circuit diagram is shown. Quantum gates are represented as boxes, except for the CNOT gate which is represented as a circle. The circuit flows from left to right. Together, the quantum gates build up a quantum circuit that performs a specific operation. [18]

Interference is when two or more quantum waves superimpose. It is said to be constructive when the combined wave increases in amplitude and destructive when the combined wave decreases in amplitude. Because quantum systems exist in a superposition of states, there can be interference between the states.

A quantum algorithm includes the specific quantum gates used to carry out an operation. Each algorithm uses a set of quantum gates that create specific interference patterns in the superposition of states from the qubits. The interference results in some qubit states being amplified and some being canceled out. The states that are amplified are taken to be the solution to the operation. When these states are measured, the qubit wave function collapses and the quantum superposition of states turns into a classical measurement of a 1 or 0. This measurement process is shown in fig. 5. Because a qubit's superposition describes the probability of a certain measurement to be true, the quantum circuit would have to run many times to get a probability distribution of the most likely result.

There are several tasks that quantum computers are better suited for than classical computers [21]. These tasks include optimization problems, searching databases, and simulations. Quantum computers can

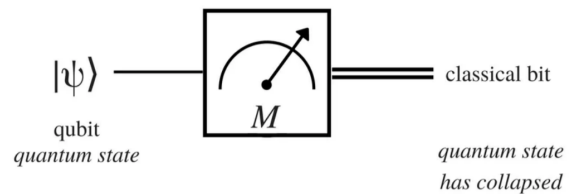


FIG. 5. When a quantum state is measured, it is no longer in a quantum state of superposition. Instead, its state collapses and a classical bit represents the measurement. [20]

better handle these problems because of their large computing power capabilities. Unlike classical computers whose power is directly related to the number of transistors, the power of a quantum computer is exponentially related to the number of qubits. However, there are still disadvantages to quantum computers. They must be kept at temperatures near absolute zero [22]. This is to prevent the matter that makes up the qubits from moving around too much, which can cause unwanted energy to be expelled and incorrectly flip qubit states. Even with extra precaution, quantum computers still have high error rates because it is possible for qubits to still interact with the environment [23]. Furthermore, unlike classical computers, quantum computers cannot easily check if a qubit is incorrect because measurement causes the qubit's wave function to collapse and remove its quantum properties. There has been research into new methods for error correction, but they have not been sufficient enough to solve the problem. Until these errors are fixed, quantum computers cannot be relied on for very complex problems.

IV. SUBFIELDS

While quantum computers still have to be improved to achieve quantum supremacy over classical computers, they still have important uses in several subfields. I will discuss three important subfields: quantum information science, quantum cryptography, and quantum machine learning.

A. Quantum Information Science

Information science is a field that studies how information is created, managed, stored, and used [24]. It is an interdisciplinary field that draws upon aspects of computer science, library science, and information management. Quantum information science extends from classical information science, but it transmits and processes information in more efficient ways using mechanisms from quantum mechanics [25].

Shannon's theory provides a foundation for information theory [26]. Shannon's 1948 paper discussed two theorems, the first being the source coding theorem and the second being the channel coding theorem. The source coding theorem describes how much data can be compressed while still being able to be restored to its original state. The channel coding theorem discusses the reliability of a channel communicating accurate bits when the channel is subject to noise and data corruption. In quantum information theory, these theories have been extended to apply to quantum systems in the form of quantum source coding and retrieval of information.

Quantum source coding is a branch of quantum information theory that deals with the compression and decompression of quantum states. To describe source coding, we must talk about the randomness of a quantum

state, or the von Neumann entropy. This is a measure of the lack of information about the state of a quantum system. The main theorem in quantum source coding says that a quantum state can be compressed or decompressed if its coding rate, or the rate at which the quantum information is compressed, exceeds the von Neumann entropy. Effective compression and decompression allow for efficient quantum communication and optimized processes.

Quantum information theory also deals with the retrieval of accessible information in quantum systems. This is the amount of classical information that can be extracted from the quantum system. The retrieval depends on the classical capacity of a quantum channel. This is the maximum rate classical data can be sent error-free in a quantum computer, and it is important because it provides a limit for information processing in a quantum computer.

The last topic of quantum information theory I will discuss is quantum teleportation. Quantum teleportation is a communication protocol that allows the state of a qubit to be known at another location without physically transferring the qubit [27]. The reason why a qubit's information is transferred indirectly is that if it was directly measured, the state would change and it would no longer be a quantum superposition. The teleportation protocol uses an entangled qubit pair for a sender and receiver. The sender of the information entangles an additional qubit with their qubit using a specific operation. The receiver then performs the sender's operation on their qubit. This "teleports" the state of the qubit from the sender to the receiver. A diagram showing quantum teleportation is shown in fig. 6. Quantum teleportation is also important because it provides security when transmitting information across a quantum system.

B. Quantum Cryptography

Cryptography is the development of secure communication channels in the presence of adversarial behavior [28]. Quantum cryptography aims to use the properties of quantum mechanics for secure channels.

RSA (Rivest–Shamir–Adleman) is a widely used encryption algorithm. It involves the use of public and private keys. The private key includes two very large prime numbers. The public key includes the product of these prime numbers and an exponent. The difficulty in the decryption without the private key lies in the fact that it is extremely time-intensive to factor a very large number into its prime factors. However, in 1994, Peter Shor published an algorithm that allowed quantum computers to factor large numbers relatively quickly and efficiently. This algorithm is shown in fig. 7 as a quantum circuit diagram. In theory, Shor's theorem would be able to break RSA cryptography. For this to happen, quantum computers would still need a large number of qubits and low error rates. While quantum computers are still being improved to reach this stage, Shor's theorem still poses a

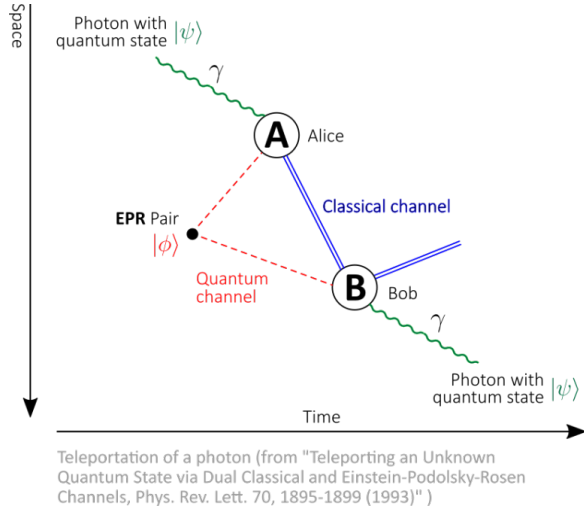


FIG. 6. The protocol of quantum teleportation is shown. There are two entangled qubits: one for Alice and one for Bob. When Alice performs an operation on her qubit and tells Bob the operation used via the communication channel, Bob can perform the same operation on his qubit and determine the original state from Alice. [18]

threat to the future of public-private key cryptography.

To combat this concern, ideas for quantum cryptography were developed. A well-known method of quantum cryptography is quantum key distribution (QKD). QKD creates secure keys that are resilient to decryption from quantum computers [30]. It works through the interaction of a sender (Alice) and a receiver (Bob) [31]. Alice and Bob establish a classical communication channel between themselves. Alice prepares a stream of photonic qubits that are entangled and then sends these qubits to Bob via a fiber optic cable. Bob measures the qubits and records their state. Alice and Bob then compare their measurements via their established communication channel. The measurements should match because the qubits are entangled. If discrepancies are found, this means there was an eavesdropper (Eve), an unauthorized third party that was trying to intercept the communication between Alice and Bob. After Eve observes the

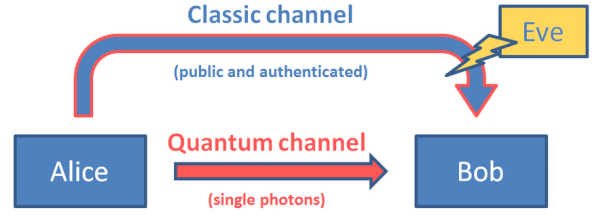


FIG. 8. A diagram describing QKD is shown. The private key is developed through the use of both a quantum channel and a classic channel between Alice and Bob. [32]

qubits, their states change, causing Bob to have different measurements than Alice. If there was an eavesdropper, Alice and Bob could reconcile each other's differences in measurement, and in turn, generate a secret private key that Eve does not know about. A diagram of QKD is shown in fig. 8.

QKD has been demonstrated to work, but it is not yet widely used. Sending photons through a fiber optic cable one photon at a time is incredibly tedious. Also, it is difficult to send qubits over long distances without being disturbed. There is current research being done to improve the capabilities of QKD, but as for now, classical public-private key cryptography remains the norm. Because of the limitations of QKD, several agencies such as the NSA have instead promoted the use of quantum-resistant algorithms [33]. These algorithms are still being developed, but they would be able to be implemented on existing platforms and provide a way for the private-public key standard to be protected from quantum computers.

C. Quantum Machine Learning

Quantum machine learning combines both quantum computing and machine learning. Its applications include improving machine learning algorithms and quantum experiments [34]. On the surface level, quantum computers can speed up machine learning processes due to their ability to compute in parallel and their ability to create logical circuits based on interference [35]. Quantum computers can also provide new methods of machine learning that have the potential to outperform the corresponding classical machine learning models. Examples of these include quantum clustering and quantum autoencoders.

Quantum clustering uses principles from quantum mechanics to group similar data points. It is based on the Schrödinger equation which describes a potential energy function. Data points are put into groups, or clusters, based on a gradient descent of the potential function. Therefore, the points are categorized based on minimizing their potential energy. This mechanism is used because it is a quantum mechanics principle that particles will want to be in their lowest energy state. Newer methods of quantum clustering include a probabilistic framework that measures how well clusters fit the data based on a likelihood function.

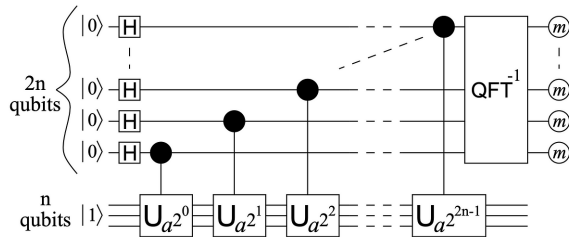


FIG. 7. The quantum circuit that makes up Shor's algorithm is shown. The algorithm consists of a classical part and a quantum part. The classical part involves repeated squaring. The quantum part involves a quantum Fourier transform (QFT). [29]

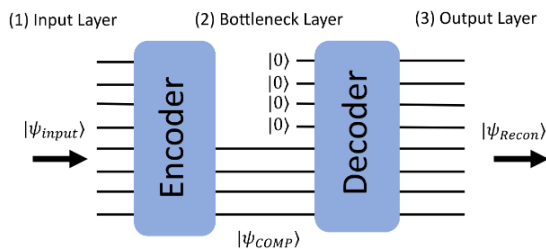


FIG. 9. The layers of a quantum autoencoder are shown. The bottleneck layer contains the compressed state while the input and output layers hold the original state. [36]

Quantum autoencoders are neural networks that can compress and reconstruct data [36]. They consist of three layers. The first layer is the input layer and also the layer that is compressed. The second layer is the bottleneck layer, and it holds the compressed state. To get to this layer, the input goes through a parameterized circuit that compresses the state to a smaller number of qubits. The final layer reconstructs the input by adding qubits via another parameterized circuit. A diagram of an autoencoder is shown in fig. 9.

Machine learning can also be used in quantum environments. An example of this is quantum reinforcement learning. This field aims to create quantum agents that can interact and learn from the world. Because the agents are based on quantum mechanics, they can use parallel computing to discover multiple possibilities for a problem. They can also utilize entanglement to more efficiently execute preferable options.

V. FUTURE

Quantum computing has made large strides in the last decade. More advances will be made in the future as the technology becomes more integrated into systems across the world. In regards to topics already discussed, there is ongoing research into better error correction mechanisms for quantum computers, better cryptography that is resistant to quantum computers, and higher quality and quantity of qubits in a quantum computer [37]. The addition of more qubits is difficult due to costs, and it is

difficult to control as more qubits would need to be super-cooled and not interfere with each other. Error correction overhead is likely to increase in quantum computers to adapt to the increased number of qubits and the need for a reliable error correction system. Hardware also needs to be improved to increase the fidelity of qubits.

While quantum computing can speed up specific processes, it is not ideal for all computer applications. Most everyday tasks are efficiently managed by classical computers. In the future, developments in quantum computing will be more focused on what will improve quantum computers in the areas where they can be superior to classical computers. These areas include problems that have the potential for a super-polynomial speedup [38]. These are problems where a quantum computer would only take time T to perform a calculation while a classical computer would take time T^2 . The Shor's algorithm is an example of a super-polynomial speedup. More research has to be done to determine other problems where this speedup can be applied.

VI. CONCLUSION

At the foundation, quantum computing is built around qubits. The quantum properties of qubits, such as superposition and entanglement, are utilized to build quantum circuits that perform calculations on a quantum computer. Quantum computing can be applied to quantum information science, quantum cryptography, and quantum machine learning. These subfields use the principles of quantum computing to perform tasks that were previously only done using classical computers. While classical computers still perform everyday tasks relatively efficiently, quantum computers are making large strides to become more efficient and solve new problems.

VII. ACKNOWLEDGEMENTS

I would like to acknowledge Kassia Schraufnagel and Sophia Figura for their recommendation for me to research more into quantum computing. I would also like to acknowledge Dr. Joseph Anderson for his expertise on how to write a scientific paper.

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